

MFML EXAM CHEAT SHEET

Last Minute Revision: Algorithms, Decompositions, and Optimization

1. VECTOR SPACE BASICS

Key Definitions

- **Span:** The set of all linear combinations.
- **Basis:** Linearly Independent set that Spans V .
- **Dimension:** Number of vectors in a Basis.

Testing Independence & Basis

To check a set of vectors $\{v_1, \dots, v_k\}$:

1. Put vectors as **columns** in matrix A .
2. Gaussian Elimination $\rightarrow$ Row Echelon Form (REF).

Interpretation:

- **Independent:** Every column has a pivot.
- **Dependent:** At least one column has NO pivot.
- **Rank(A):** Count of non-zero rows (pivots).

System Consistency ($Ax = b$)

Form Augmented Matrix $[A|b]$ and reduce to REF.

- **No Solution:** $\text{Rank}(A) < \text{Rank}(A|b)$ (Pivot in last col).
- **Unique:** $\text{Rank}(A) = \text{Rank}(A|b) = \text{Total Variables}$.
- **Infinite:** $\text{Rank}(A) = \text{Rank}(A|b) < \text{Total Variables}$.

Gram-Schmidt Process

Goal: Turn Basis $\{x_1, x_2\}$ into Orthonormal $\{e_1, e_2\}$.

1. **First Vector:** Normalize x_1 .

$$e_1 = \frac{x_1}{\|x_1\|}$$

2. **Second Vector:** Remove projection of x_2 on e_1 .

$$v_2 = x_2 - \langle x_2, e_1 \rangle e_1$$

3. **Normalize:**

$$e_2 = \frac{v_2}{\|v_2\|}$$

4. **General Step:** $v_k = x_k - \sum_{j=1}^{k-1} \langle x_k, e_j \rangle e_j$

2. MATRIX DECOMPOSITIONS

Eigen Decomposition (Diagonalization)

Constraint: Square Matrix ($n \times n$)

$$A = PDP^{-1}$$

- **1. Find Eigenvalues (λ):** Solve $\det(A - \lambda I) = 0$.
- **2. Find Eigenvectors (v):** Solve $(A - \lambda I)v = 0$.
- **3. Construct:** $P = [v_1 \dots v_n]$, $D = \text{diag}(\lambda_i)$.

Singular Value Decomposition (SVD)

Constraint: Any Matrix ($m \times n$)

$$A = U\Sigma V^T$$

Standard Strategy (for Tall Matrix $m \geq n$):

- **1. Compute $A^T A$:** (Smallest square matrix).
- **2. Find Right Singular Vectors (V):** Eigenvectors of $A^T A$.
- **3. Singular Values (Σ):** $\sigma_i = \sqrt{\lambda_i(A^T A)}$.
- **4. Find Left Singular Vectors (U):**

$$u_i = \frac{1}{\sigma_i} A v_i$$

Reverse Strategy (for Fat Matrix $n > m$):

- **1. Compute AA^T :** (Smallest square matrix).
- **2. Find Left Singular Vectors (U):** Eigenvectors of AA^T .
- **3. Find Right Singular Vectors (V):**

$$v_i = \frac{1}{\sigma_i} A^T u_i$$

Cholesky Decomposition

Constraint: Symmetric, Positive-Definite

$$A = LL^T \quad (L \text{ is Lower Triangular})$$

Procedure (3×3):

- $l_{11} = \sqrt{a_{11}}$
- $l_{21} = \frac{a_{21}}{l_{11}}, \quad l_{31} = \frac{a_{31}}{l_{11}}$
- $l_{22} = \sqrt{a_{22} - l_{21}^2}$
- $l_{32} = \frac{a_{32} - l_{31}l_{21}}{l_{22}}$
- $l_{33} = \sqrt{a_{33} - (l_{31}^2 + l_{32}^2)}$

3. CALCULUS & OPTIMIZATION

Gradients & Jacobian

- **Gradient (∇f):** Vector of partials $\frac{\partial f}{\partial x_i}$.
- **Jacobian (J):** Matrix of partials. $J_{ij} = \frac{\partial f_i}{\partial x_j}$.

Optimization (The Hessian Test)

Goal: Find max/min of $f(x, y)$.

1. **Critical Points:** Solve $\nabla f = 0 \implies f_x = 0, f_y = 0$.
2. **Hessian Matrix:**

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

3. **Classify** ($D = \det(H) = f_{xx}f_{yy} - f_{xy}^2$):
 - **Local Min:** $D > 0$ AND $f_{xx} > 0$.
 - **Local Max:** $D > 0$ AND $f_{xx} < 0$.
 - **Saddle Point:** $D < 0$.
 - **Inconclusive:** $D = 0$.

Backpropagation (Chain Rule)

To find gradient of Loss L w.r.t parameter θ_i :

$$\frac{\partial L}{\partial \theta_i} = \frac{\partial L}{\partial f_K} \cdot \frac{\partial f_K}{\partial f_{K-1}} \cdots \frac{\partial f_{i+1}}{\partial \theta_i}$$

Linearization

Approximate $f(x)$ by a tangent plane near point a :

$$L(x) = f(a) + \nabla f(a)^T (x - a)$$

4. TAYLOR SERIES EXPANSION

A. Univariate (1 Variable)

Approximation of $f(x)$ near $x = a$:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3$$

Maclaurin Series: Set $a = 0$.

B. Multivariate (2 Variables)

Expansion of $f(x,y)$ near point (a,b) . Let $\mathbf{h} = (x - a)$ and $\mathbf{k} = (y - b)$.

1. Linear Term (Degree 1):

$$f(a,b) + [f_x(a,b)h + f_y(a,b)k]$$

2. Quadratic Term (Degree 2):

$$+ \frac{1}{2!} [f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2]$$

3. Cubic Term (Degree 3):

$$+ \frac{1}{3!} [f_{xxx}h^3 + 3f_{xxy}h^2k + 3f_{xyy}hk^2 + f_{yyy}k^3]$$

Note: All partials evaluated at (a,b) .

C. General Vector Form

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a})^T (\mathbf{x} - \mathbf{a}) + \frac{1}{2} (\mathbf{x} - \mathbf{a})^T H (\mathbf{x} - \mathbf{a})$$

5. QUICK IDENTITY CHEAT SHEET

Vector Calculus Identities

- $\nabla_x (b^T x) = b$
- $\nabla_x (x^T x) = 2x$
- $\nabla_x (x^T A x) = (A + A^T)x$
- $\nabla_x (x^T A x) = 2Ax$ (if A is Symmetric)

Inner Products & Geometry

- **Dot Product:** $\langle x, y \rangle = x^T y$.
- **Norm:** $\|x\| = \sqrt{\langle x, x \rangle}$.
- **Angle θ :** $\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|}$.
- **Orthogonal:** $\langle x, y \rangle = 0$.

6. THE "SMART STUDENT" STRATEGY

1. **The Smaller Matrix Rule:** For SVD, always compute the smaller of $A^T A$ or AA^T first.
 - Tall Matrix $(3 \times 2)? \rightarrow$ Compute $A^T A$ (2×2) .
 - Fat Matrix $(2 \times 3)? \rightarrow$ Compute AA^T (2×2) .

2. **Optimization Check:** For Max/Min, checking $\det(H) > 0$ is not enough. You **must** check the sign of f_{xx} !
3. **Independence Shortcut:** If you have more vectors than dimensions (e.g., 4 vectors in $\mathbb{R}^3$), they are **Dependent**. Don't waste time calculating.